

Preparing customer data for churn classification

Midterm project proposal

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Course / section: ITAI-1374

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Dataset and objective

This project will use the Telco Customer Churn dataset published by BlastChar on Kaggle. It contains 7,043 customer records and 21 columns describing service subscriptions, tenure, contracts, billing, and demographic attributes. The target, Churn, indicates whether a customer left within the last month. The dataset is an IBM educational sample. Our objective is to prepare reproducible training data for a later binary classification project that studies which customer characteristics are associated with churn.

Preparation and exploratory analysis

We will preserve the original CSV and use Python to create a reproducible, stratified 70% training / 30% testing split. All EDA and preprocessing decisions will use training data. The testing partition will remain unprocessed and will not be explored or used to choose transformations during the midterm.

We will inspect types, missing entries, duplicate identifiers, category labels, numeric ranges, and class balance. Blank or invalid numeric text in TotalCharges will become missing values and be filled using the training median, with an indicator recording missingness. Categorical variables will be one-hot encoded and Churn mapped to 0/1. Numeric predictors will be scaled, and a square-root transformation will demonstrate how a skewed distribution changes. Fixed-rule features will summarize service subscriptions and customer tenure. Customer identifiers will be excluded from model inputs. Random oversampling will demonstrate class balancing in a separate training copy while preserving the original training class proportions in the main cleaned dataset.

Evidence and deliverables

The Jupyter notebook will show training samples and Matplotlib plots before and after preparation, explain each decision, and check for missing or non-finite model inputs. Deliverables will include the source URL document, this proposal, the executed notebook, cleaned training data, and a reflection with each member's actual contribution. Original data and the untouched holdout will remain separate.

Future machine learning use

The final project could compare logistic regression, decision tree, and random forest classifiers using stratified cross-validation within the training partition. Learned preprocessing and any resampling will be fitted inside each training fold. Evaluation will consider precision, recall, F1, ROC-AUC, and precision-recall performance because accuracy can conceal class imbalance. After selecting the workflow, the saved holdout will be transformed using training-fitted steps and evaluated once. The sample supports educational comparison; associations will not be treated as proof of causation or guaranteed future behavior.

Source: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>